Starter application and tests

Vasisht Gaddale | Coupon Validator | Assignment 31

The CLI runs used save-20, 199,49.5,320 and 2024-01-01 as the valid inputs. The command panels are rendered from captured terminal output.

Original application with valid input

```
$ python coupon_validator.py
==== Running split_code ====
Enter a coupon code (e.g. save-20): Prefix and Value: ('SAVE', 20)

==== Running most_expensive ====
Enter item prices as CSV (e.g. 199,49.5,320): Most Expensive Item: 320.0

==== Running expiry_days ====
Enter an expiry date as YYYY-MM-DD: Days Since 2024: 1
```

Hypothesis calendar test against the original code

```
$ python -m pytest -q --tb=short test_coupon_validator.py -k counts_calendar_days
F                                                                        [100%]
================================== FAILURES ===================================
_____________________ test_expiry_days_counts_calendar_days _____________________
test_coupon_validator.py:58: in test_expiry_days_counts_calendar_days
    def test_expiry_days_counts_calendar_days(offset):
                   ^^^
test_coupon_validator.py:60: in test_expiry_days_counts_calendar_days
    assert expiry_days(expiry.isoformat()) == offset + 1
E   AssertionError: assert -4 == (-1 + 1)
E    +  where -4 = expiry_days('2023-12-31')
E    +    where '2023-12-31' = <built-in method isoformat of datetime.date object at
0x7f2ca1c34990>()
E    +      where <built-in method isoformat of datetime.date object at 0x7f2ca1c34990> =
datetime.date(2023, 12, 31).isoformat
E   Failing test case: test_expiry_days_counts_calendar_days(
E       offset=-1,
E   )
========================== short test summary info ============================
FAILED test_coupon_validator.py::test_expiry_days_counts_calendar_days - Asse...
1 failed, 8 deselected in 0.24s
```

Hypothesis found a valid date where the original arithmetic disagrees with the calendar. December 31, 2023 should be day 0 under the day-1 convention. The original function returned -4.

Fixed application and tests

Vasisht Gaddale | Coupon Validator | Assignment 31

Invalid inputs return None. Coupon prefixes are normalized to uppercase, cart prices must be finite and nonnegative, and expiry offsets use real calendar dates with 2024-01-01 as day 1.

Fixed application with the same valid input

```
$ python coupon_validator.py
==== Running split_code ====
Enter a coupon code (e.g. save-20): Prefix and Value: ('SAVE', 20)
==== Running most_expensive ====
Enter item prices as CSV (e.g. 199,49.5,320): Most Expensive Item: 320.0
==== Running expiry_days ====
Enter an expiry date as YYYY-MM-DD: Days Since 2024: 1
```

Fixed application with invalid input

```
$ python coupon_validator.py
==== Running split_code ====
Enter a coupon code (e.g. save-20): Prefix and Value: Invalid input
==== Running most_expensive ====
Enter item prices as CSV (e.g. 199,49.5,320): Most Expensive Item: Invalid input
==== Running expiry_days ====
Enter an expiry date as YYYY-MM-DD: Days Since 2024: Invalid input
```

Completed Hypothesis suite

```
$ python -m pytest -q test_coupon_validator.py
.........                                                                [100%]
9 passed in 0.55s
```

Nine properties cover mixed-type inputs, valid results and invalid-input rejection across the three functions.